

A CNN-LSTM APPROACH FOR ACCURATE DROWSINESS AND DISTRACTION DETECTION IN DRIVERS

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ABSTRACT. *In this study, we propose a Convolutional Neural Network – Long Short-Term Memory (CNN-LSTM) approach for accurate drowsiness and distraction detection in drivers. Our research aims to develop a robust system capable of effectively identifying drowsiness and distractions in drivers. We adopt the FaceMesh algorithm to extract facial landmarks, explicitly focusing on eye and mouth regions. The aspect ratio of these facial features is calculated to detect potential signs of drowsiness. Additionally, we introduce the Intersection over Union (IOU) technique to identify instances where the hand intersects with the mouth, indicating that the hand is raised to cover the mouth when yawning. Head pose estimation is employed to ascertain the subject's gaze direction and detect driver inattention. To integrate and analyze these features effectively, we use a CNN-LSTM model. To evaluate the performance of our approach, we conduct experiments on a dataset comprising 65 videos, each approximately 1 minute in length. Remarkably, our model achieves 93.60%, 93.61%, 91.92%, and 92.68% of accuracy, precision, recall, and F1-score, respectively, in accurately detecting drowsiness and distractions in drivers. These results highlight the potential of our proposed approach in enhancing driver safety and preventing accidents caused by fatigue and distractions.*

Keywords: Drowsiness detection, Deep learning, CNN-LSTM, Risky driver, Distracted driving

1. Introduction. One of the leading causes of driver unpreparedness is driver drowsiness, which poses a danger to oneself and other road users. According to road accident statistics in Thailand, the department of disease control stated that about 4% of accidents are caused by drivers experiencing drowsiness or falling asleep at the wheel [1]. Globally, drowsy driving is estimated to be a factor in 20% of road accidents [2]. This underscores the importance of addressing and mitigating the issue. Anticipating and alerting drivers before they become drowsy could potentially mitigate the severity of accidents.

With the rising number of vehicles on roads and the growing concern over road safety, detecting risky driving behaviour has become crucial. According to the World Health Organization (WHO) [2], approximately 1.35 million people die annually from road accidents, most of which are caused by risky driving behaviour such as speeding, distracted driving, and aggressive driving. As a result, developing effective methods for detecting and preventing such behaviour on the road is imperative. Machine Learning (ML) techniques have emerged as powerful tools for detecting risky driving behaviour in recent years. By analyzing large amounts of data collected from sensors and cameras installed

in vehicles, ML algorithms can learn to recognize behaviour patterns indicative of risky driving. These algorithms can then generate real-time alerts and notifications for drivers, reducing the likelihood of accidents [3].

This paper provides a thorough examination of state-of-the-art Machine Learning (ML) techniques employed in the detection of risky driving behavior. We explore a spectrum of dangerous driving behaviors, delving into the challenges associated with their identification. In addition to the conventional discussion, we scrutinize various data sources, including vehicle sensors [4], thermal cameras [5], EOG sensors [6], and EEG sensors [7, 8], offering a multifaceted view of the landscape.

Within the realm of ML algorithms for dangerous driving detection, we specifically employ CNN-LSTM to develop our system, enhancing the sophistication of our approach. Our academic contributions encompass a comprehensive review of existing literature and a meticulous evaluation of the proposed algorithm's performance [9, 10]. This dual focus ensures a robust foundation for understanding the state-of-the-art and the efficacy of our method.

Moving beyond the academic sphere, our paper extends its impact to practical applications. We proudly present real-world applications derived from our research, highlighting the tangible contributions of our work to the field. These applications underscore the relevance and applicability of our findings in addressing challenges and enhancing safety measures in the domain of risky driving.

In the final section, we showcase our experimental results, rigorously evaluating the performance of different algorithms on our dataset. Through this analysis, we not only discuss the limitations inherent in the current methodologies but also chart potential future directions for this evolving field. In essence, our paper offers a comprehensive overview of the current state-of-the-art in ML-driven risky driving detection, encompassing academic depth, algorithmic sophistication, and practical applicability, while paving the way for further research [4, 5, 6, 7, 8, 9, 10].

We outline the structure of the remaining sections of the paper. Section 2 reviews the existing literature on state-of-the-art fall detection. Section 3 introduces our ideas and methods for vision-based fall detection. Section 4 presents the results of the work and discusses its findings. Finally, Section 5 summarizes the main points and discusses future research directions.

2. Related Works. This section will explore the relevant literature and research on risky driving behaviours. It is essential to have a thorough understanding of this area to shape our research and contributions effectively. As technology and machine learning continue to advance at breakneck speeds, researchers and industry experts emphasize studying risky driving behaviours. This section aims to provide a detailed overview of the current research and technological advancements that are helping us better comprehend dangerous driving and how it can be detected. This comprehensive literature review categorizes existing approaches into three distinct aspects: Biological-based, Vehicle-based, and Vision-based.

Biological-based approaches: Biological-based methods, encompassing physiological signals such as Electroencephalogram (EEG) [11, 12], Electrocardiogram (ECG) [13], and Electrophysiological (EOG) [14] have been prominent in early-stage drowsiness detection systems. Prior research, such as that conducted by [15], has demonstrated the correlation between brain and heart activity changes and driver drowsiness. Despite promising outcomes, the practical implementation of these methods takes time and effort. Physiological sensors can be intrusive and may induce discomfort, leading to potential user reluctance and affecting the system's overall reliability. Moreover, the accuracy of these systems can be influenced by external factors such as environmental conditions and individual variations, limiting their robustness across diverse driving scenarios.

Vehicle-based approaches: Vehicle-based approaches rely on data derived directly from the vehicle, including steering patterns, speed, and lane-keeping behaviour. Prior studies, such as that by [16], have explored the use of vehicle dynamics for drowsiness detection. While these methods offer non-intrusive means of monitoring driver behaviour, they often exhibit limitations in differentiating between distraction and drowsiness. The dependence on vehicle dynamics alone can lead to false positives or negatives, mainly when external stimuli rather than internal states influence driver actions. Moreover, real-world driving conditions, such as traffic congestion, can introduce noise into the data, impacting the system's reliability.

Vision-based approaches: Transitioning from the drawbacks of Biological and Vehicle-based methods to Vision-based approaches introduces a promising alternative, especially in capturing nuanced driver behaviour. Vision-based systems, utilizing cameras to analyze facial features, eye movements, and overall driver behaviours [17, 18], have effectively detected drowsiness and distraction [19]. However, challenges persist, including potential limitations in capturing subtle facial expressions, particularly in low-light conditions. The need for robust image processing algorithms becomes crucial in overcoming these challenges. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNN) [17, 20] and Long Short-Term Memory (LSTM) networks [21], have opened avenues for addressing these concerns, offering a more accurate and versatile solution for drowsiness and distraction detection.

Rationale for Vision-based approaches: The drawbacks of Biological and Vehicle-based methods underscore the necessity for a more comprehensive approach. The visual modality provides a rich information source that can address other methods' limitations. By leveraging a CNN-LSTM architecture, our proposed approach aims to harness the power of deep learning to extract meaningful features from visual data. This approach aligns with recent findings by [22], showcasing the efficacy of deep learning models in capturing intricate patterns related to driver behaviour [23, 24]. Integrating temporal dependencies through LSTM enhances the model's ability to discern subtle changes in facial expressions and eye movements. This literature review sets the stage for exploring and validating our CNN-LSTM model within the broader context of driver safety systems.

3. Proposed Method. Based on a review of research documents concerning drowsiness detection and forward gaze avoidance during driving, even though the detection of drowsiness can yield high accuracy using direct physiological signal measurement methods or behavioural patterns associated with sleepiness, such as Electroencephalography (EEG), Electrooculography (EOG), heart rate monitors, and infrared sensors, the placement of sensors has an impact on accuracy. Prolonged sensor attachment may lead to discomfort, unease, or potential injury, and over an extended period, wearers may not cooperate with the sensor attachment process. Consequently, in this study, the approach of computer vision is adopted, utilizing RGB cameras situated in the vicinity. The proposed methodology employs computer vision techniques, explicitly using Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Intersection over Union (IOU) between mouth and hand, and HeadPose, derived from Face and Hand Landmarks extracted using deep learning libraries, specifically FaceMesh and Hand Landmarks from MediaPipe. The data is then organized into temporal sequences (Time Series) for training a CNN-LSTM hybrid model as shown in Figure 5. The research methodology is delineated into five sequential stages, as depicted in Figure 1(a).

3.1. Datasets. We conducted a research project that involved the use of a self-recorded dataset. The dataset consists of 65 videos, with 18 videos being male-video and the remaining videos being female-video, each lasting about 1 minute, and was captured from male and female volunteers as shown in Figure 2. The videos contained instances of three

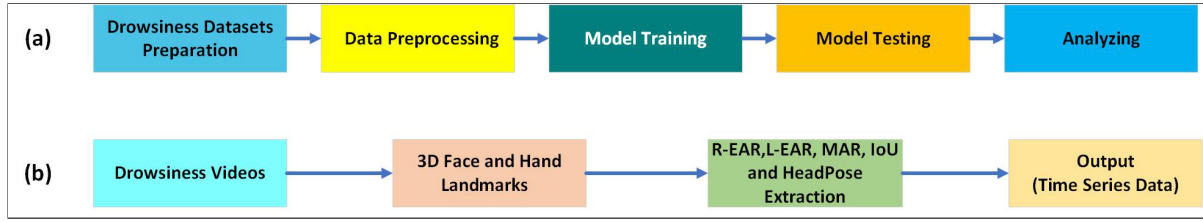


FIGURE 1. (a) An overview of the proposed method; (b) the data preparation step

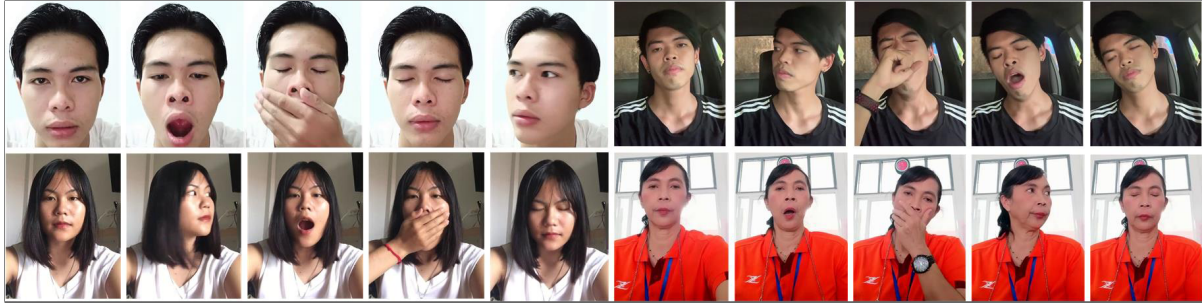


FIGURE 2. Samples of dataset

distinct states, namely “Normal”, “Drowsy”, and “Inattention”. The video dataset was uniformly processed to achieve a consistent temporal sequence with a frame rate of 24 frames per second. Also, the videos were resized to a standard width of 540 pixels to facilitate consistent feature extraction. The next step was to label the activities of individuals in the videos through data labelling. The dataset was divided into two subsets, with 70% allocated for model training and 30% reserved for model evaluation in subsequent experiments.

3.2. Data preprocessing. To prepare the data for our research, we undertook two primary stages. Firstly, we extracted three-dimensional data points on the face and hands using FaceMesh and Hand Landmark from the MediaPipe library. Secondly, we computed vital metrics such as Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Intersection over Union (IOU) between the mouth and hands, and HeadPose. These metrics were structured as time series data to serve as input for the subsequent Convolutional Neural Network – Long Short-Term Memory (CNN-LSTM) model, as depicted in Figure 1(a).

1) Three-Dimensional Facial and Hand Landmarks (3D Face and Hand Landmarks): The images undergo facial detection as a preliminary step to obtain precise and comprehensive information regarding the Facial and Hand Landmarks in three dimensions. Subsequently, 468 three-dimensional data points are extracted from the face, utilizing the highly reliable and precise FaceMesh technology as shown in Figure 4(a). In addition, the hand landmarks are also carefully delineated and consist of 21 points as shown in Figure 4(b). This is achieved by employing the Hand Landmarks feature from MediaPipe, an open-source framework developed by Google. This framework is specifically designed to offer customizable, cross-platform machine learning solutions that are highly suitable for real-time and streaming applications, and it is renowned for its speed and efficiency.

2) Computation of Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Intersection over Union (IOU) between mouth and hand, and HeadPose: In this section, we outline the computation process for several key metrics, including Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Intersection over Union (IOU) between the mouth and hands, and HeadPose, based on the 468 Facial Landmarks. R-EAR, L-EAR, and MAR are calculated from the points surrounding the eyes and mouth, as delineated by

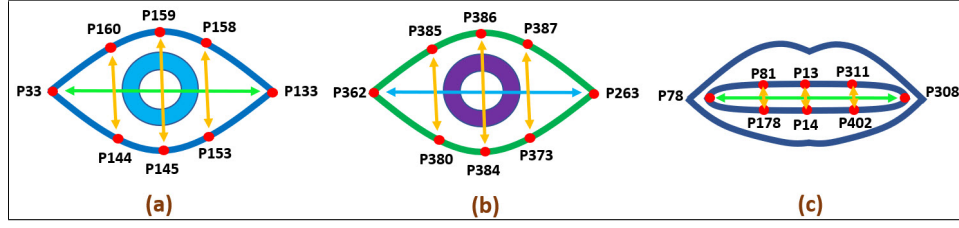


FIGURE 3. (a) Eight coordinates utilized for R-EAR calculation; (b) eight coordinates utilized for L-EAR calculation; (c) eight coordinates used in MAR calculation

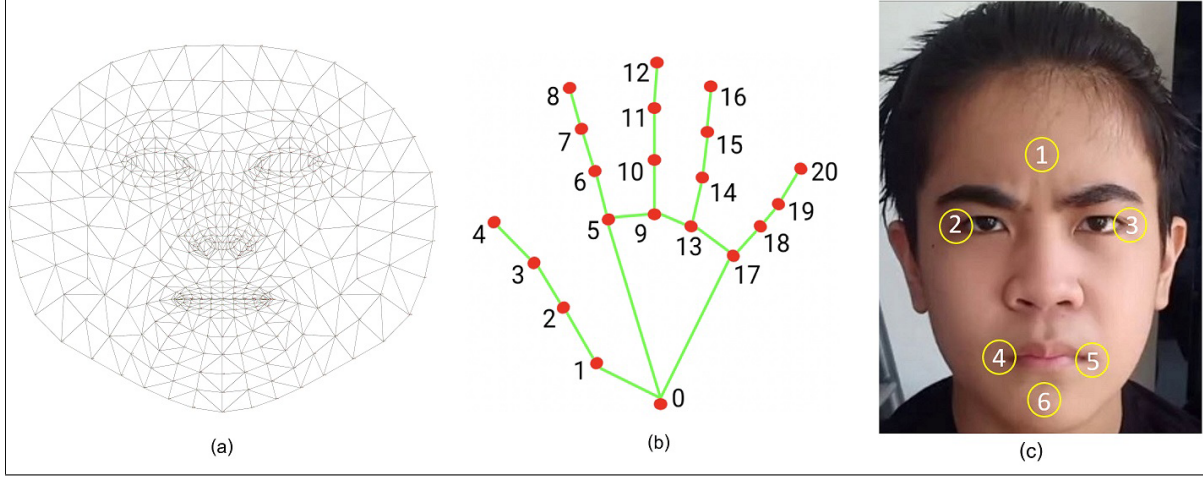


FIGURE 4. (a) Key Facial Landmarks at 468 positions obtained from the FaceMesh model [25]; (b) key Hand Landmarks at 21 positions obtained from the hand landmarks model [25]; (c) the 6 points on the subject's face

Equations (1), (2) and (3), respectively (see Figure 3 explicitly delineates the coordinates of each notation). Additionally, the Intersection over Union (IOU) between the mouth and hands is computed using Equation (4).

$$R-EAR = \frac{\|P_{160} - P_{144}\| + \|P_{159} - P_{145}\| + \|P_{158} - P_{153}\|}{3\|P_{33} - P_{133}\|} \quad (1)$$

$$L-EAR = \frac{\|P_{385} - P_{380}\| + \|P_{386} - P_{374}\| + \|P_{387} - P_{373}\|}{3\|P_{362} - P_{263}\|} \quad (2)$$

$$MAR = \frac{\|P_{81} - P_{178}\| + \|P_{13} - P_{14}\| + \|P_{311} - P_{402}\|}{3\|P_{78} - P_{308}\|} \quad (3)$$

$$IOU = \frac{Area-of-Overlap}{Area-of-Union} \quad (4)$$

HeadPose refers to the calculation of the angles of left-right rotation (ϕ) and up-down tilt (θ) from six points on the face, namely the forehead, the edge of the eyes, the chin, and the edge of the mouth, as illustrated in Figure 4(c). This calculation uses the Perspective-n-Point (PnP) method [26], as shown in Equation (5).

$$S \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & \gamma & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad (5)$$

where S represents the scale factor; γ denotes the skew parameter; (u, v) represents a 2D point in image space (in pixels); (x, y, z) corresponds to a 3D point in world coordinate

space; (u_0, v_0) signifies the image's center point (in pixels); (f_x, f_y) denotes the focal length in pixels.

The Rotation vector (R_{vec}) obtained from Equation (5) is subsequently used to compute the Rotation matrix (R_{mat}) employing Rodrigues' rotation formula [27], followed by the calculation of the angles ϕ and θ [28].

The collected data from each video is arranged in six rows, which consists of R-EAR, L-EAR, MAR, IOU, and HeadPose parameters along the x and y axes. This sequence is divided into windows of 16 frames with an overlapping configuration that shifts every eight frames. These segmented sequences are then combined with data from other videos. A random shuffling process is implemented before dividing the dataset into a training subset (70%) and a testing subset (30%) for model training and evaluation to ensure statistical accuracy and reduce data-order dependencies.

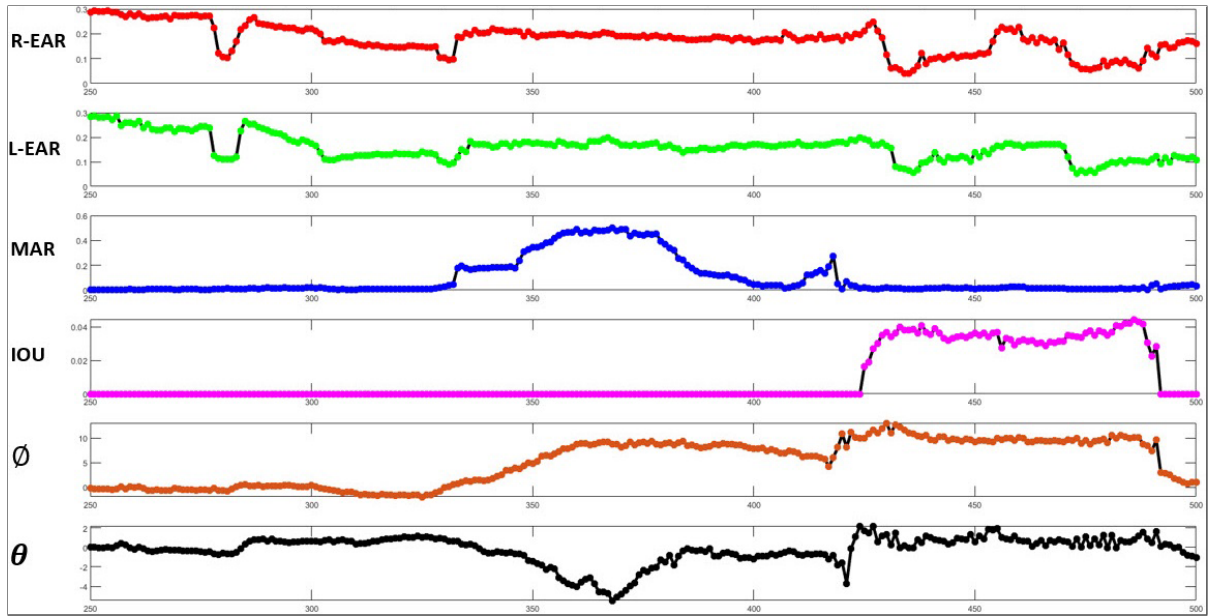


FIGURE 5. Time-series data for CNN-LSTM

3.3. Model training. After the necessary preprocessing of the train videos, the data is divided into two subsets: the Train Dataset, which comprises 70% of the data for model training, and the Test Dataset, which contains 30% of the data for model performance evaluation. The research employs a Convolutional Neural Network – Long Short-Term Memory (CNN-LSTM) deep learning architecture to analyze and recognize human activities related to drowsiness. This architecture combines Convolutional Neural Networks (CNNs) with Recurrent Neural Networks (RNNs) to interpret the temporal sequence (time series) of features and recognize various human activities. The model is designed to identify drowsiness and inattention in video footage, as depicted in Figure 6(a). The input data consists of a one-dimensional array comprising six rows and 16 frames divided into two equal segments. This data is subsequently inputted into a CNN model, whose architecture is illustrated in Figure 6(b). The CNN model comprises two layers of 1D convolution, each with 64 filters and a kernel size 3. A Dropout layer with a dropout rate of 0.5 is employed to mitigate overfitting. Then, 1D Max Pooling with a size of 2 is applied, then flattening the output. The resulting output from the CNN model is then fed into an LSTM model to capture temporal patterns in the features and identify various human activities, such as drowsiness and inattention. In order to prevent overfitting in the LSTM model, a dropout layer with a dropout rate of 0.5% is incorporated. Subsequently, the data passes through Dense layers, concluding with a Softmax layer for classification.

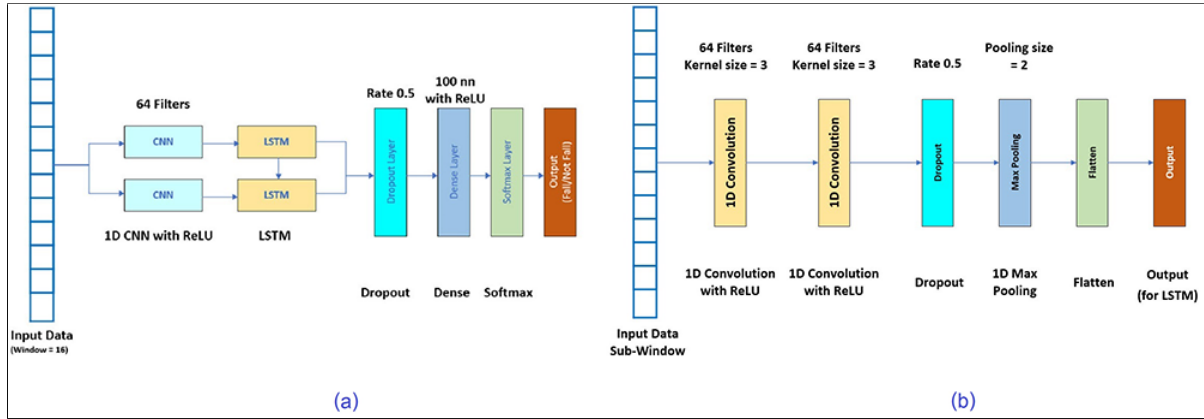


FIGURE 6. Architecture of the drowsiness detection system using CNN-LSTM

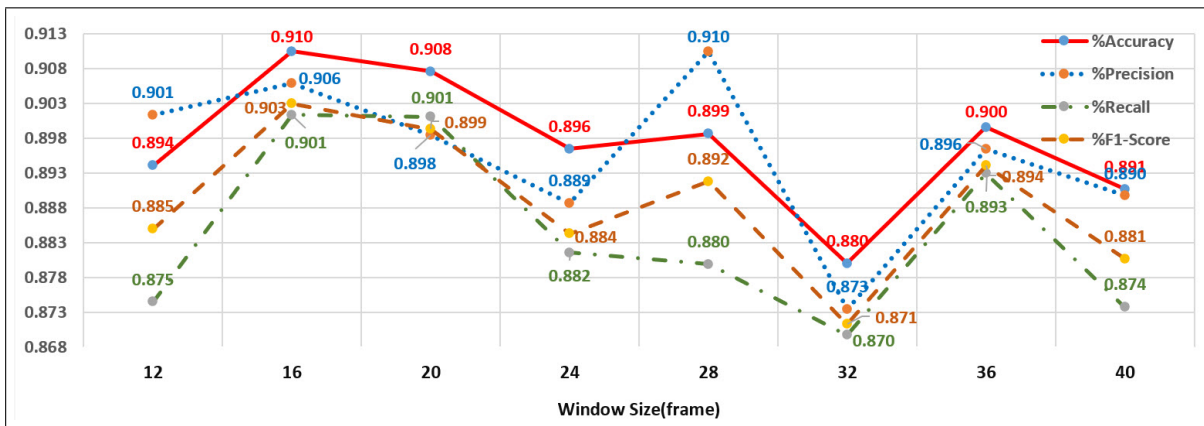


FIGURE 7. Accuracy, precision, recall and F1-score versus different window sizes

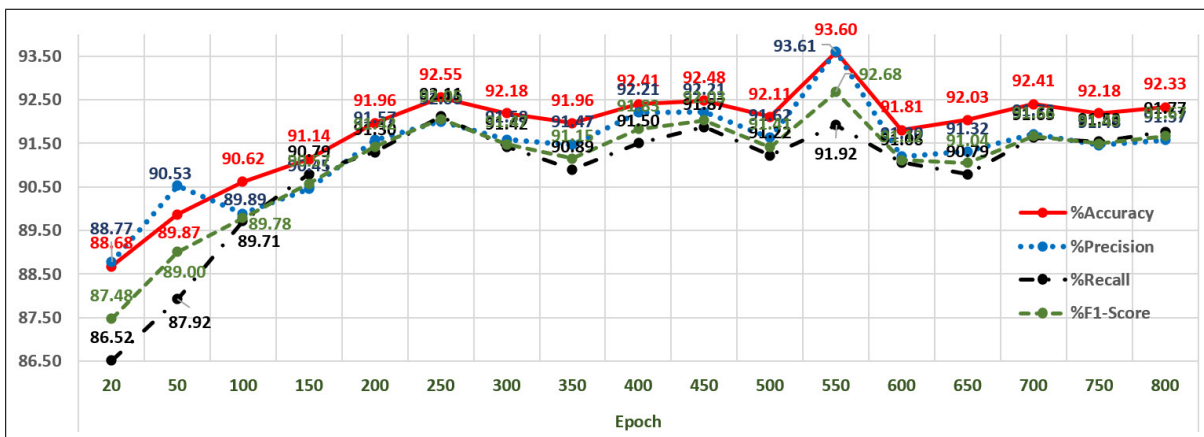


FIGURE 8. Accuracy, precision, recall and F1-score versus different epochs

The model training begins by determining an appropriate window size, teaching the model using input windows of various sizes ranging from 12 to 40 frames, and conducting 100 epochs. Figure 7 shows that the highest accuracy achieved is 91.00%, with a precision of 90.60%, a recall of 90.10%, and an F1-score of 90.30% when the window size is set to 16 frames.

Subsequently, the optimal number of epochs is determined by training the model with input data using a window size of 16 frames and varying the epochs from 20 to 800. As depicted in Figure 8, the best results are obtained when the epoch count is set to 550,

yielding an accuracy of 93.60%, precision of 93.61%, recall of 91.92%, and an F1-score of 92.68%, respectively.

3.4. Model evaluation. To evaluate the performance of our model, we trained two additional models for comparison: a CNN model and a ConvLSTM model. Both of these models used the same input data configuration, which had a window size of 16 frames and underwent training for 550 epochs. The LSTM component of these models shared the same parameters as the LSTM within our CNN-LSTM model. We then assessed these models using the same test dataset to evaluate the CNN-LSTM model. It allowed us to compare the results of all three models using four performance metrics derived from the confusion matrix: accuracy, precision, recall, and F1-score.

More detailed descriptions of the LSTM and ConvLSTM models are provided below.

LSTM, short for Long Short-Term Memory, represents a Recurrent Neural Network (RNN) architecture utilized in deep learning and artificial intelligence. LSTMs excel at processing and making predictions based on sequential data, such as time series data and natural language text, among other applications. They were introduced to overcome some limitations of traditional RNNs, which often struggle to capture long-range dependencies within sequences. Our model is defined with a single hidden LSTM and dropout layers designed to prevent overfitting the training data. Finally, a densely connected layer is employed to interpret the features extracted by the LSTM hidden layer, followed by a final output layer for making predictions.

The ConvLSTM deep learning architecture is employed for analyzing and recognizing human activities related to drowsiness. This robust neural network architecture combines CNN and LSTM models to interpret temporal sequences of features and identify various human activities. The input data is represented as a 1D sequence of six rows with a length of 16 frames (1 window) and is fed into a 2D ConvLSTM layer equipped with 64 filters and 3×3 kernel sizes. To prevent overfitting, a Dropout layer with a dropout rate of 50% is introduced. The data is then flattened and passed through Dense layers, followed by a Softmax layer for classification.

4. Performances. In our comprehensive experiments, we systematically compared the performance of the CNN-LSTM model against LSTM and ConvLSTM models to validate its effectiveness in detecting drowsiness and inattention. Utilizing our dataset, we conducted diverse experiments, the outcomes of which are outlined below.

The overall accuracy, particularly in drowsiness detection, demonstrated that the CNN-LSTM model outperformed its counterparts. Specifically, the accuracy rates were 93.6%, 93.0%, and 90.77% for CNN-LSTM, ConvLSTM, and LSTM, respectively. For drowsiness detection, the accuracy rates followed the same order: 95.62%, 93.74%, and 90.61%. When shifting the focus to inattention detection, ConvLSTM exhibited the highest accuracy at 88.93%, LSTM at 87.2%, and CNN-LSTM had the lowest accuracy at 83.97%, as illustrated in Figure 9.

To delve deeper into the evaluation, we incorporated three additional metrics, precision, recall, and F1-score, presented in Table 1. The CNN-LSTM model demonstrated superior precision and F1-score at 93.61% and 92.68%, respectively. ConvLSTM followed closely with a precision of 92.40% and an F1-score of 92.34%. LSTM lagged with the lowest precision and F1-score, at 90.96% and 90.48%, respectively. Regarding recall, ConvLSTM led with the highest value at 92.28%, then CNN-LSTM at 91.92%, and LSTM with the lowest recall value at 90.05%.

These findings strongly support the proposition that the CNN-LSTM model surpasses other deep learning architectures in effectively detecting drowsiness, showcasing significantly higher accuracy and lower error rates. Despite a slightly lower performance in detecting inattention compared to ConvLSTM and LSTM, the CNN-LSTM model

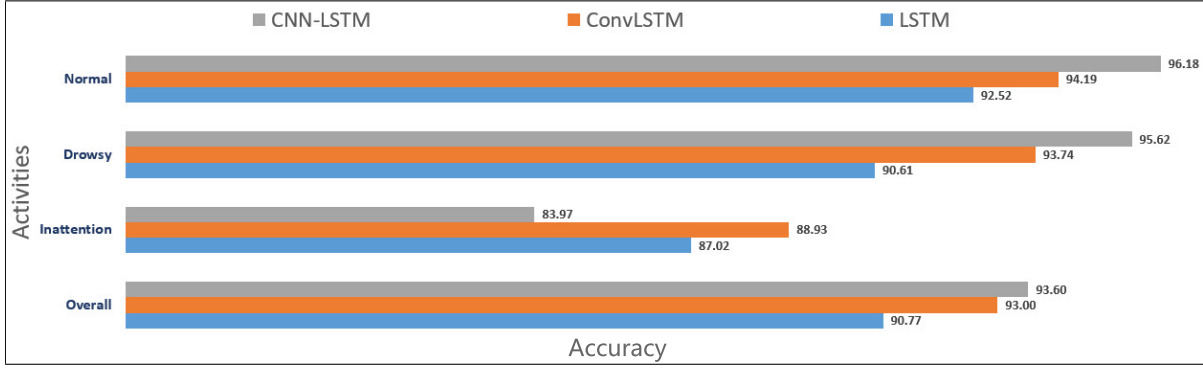


FIGURE 9. Bar chart showing accuracy of different models

TABLE 1. Comparison of the accuracy, precision, recall and F1-score for different methods (%)

Model	Precision	Recall	F1-score	Accuracy
LSTM	90.96	90.05	90.48	90.77
ConvLSTM	92.40	92.28	92.34	93.00
CNN-LSTM	93.61	91.92	92.68	93.60

consistently delivers the highest overall accuracy, precision, and recall compared to the other two models.

Upon further analysis, it becomes evident that all three models perform admirably, each showcasing strengths in different aspects. The CNN-LSTM model has the highest precision, F1-score, and accuracy. On the other hand, ConvLSTM excels in recall, highlighting its prowess in effectively capturing positive instances. These nuanced insights provide a comprehensive understanding of the relative strengths and trade-offs among the three models, aiding in an informed model selection based on specific use-case requirements.

5. Conclusion and Future Works. In this article, we employed a vision-based approach to detect driver drowsiness and inattention using Google's MediaPipe library, specifically FaceMesh and Hand Landmark, which are deep-learning components of the MediaPipe library. These components were utilized to identify the driver's eye, mouth, and hand positions to calculate six features. Subsequently, we trained a model using a combination of CNN and LSTM with a window size of 16 frames and 550 epochs. The system successfully detected driver drowsiness and inattention, achieving an accuracy of 93.60%, precision of 93.61%, recall of 91.92%, and an F1-score of 92.68%.

While this method provides high accuracy and reduces the discomfort and irritation associated with wearing physical devices, it still exhibits more errors than directly using sensors to measure body signals. Additionally, drowsiness while driving is a hazardous event that can lead to severe accidents. Therefore, combining this method with other approaches to enhance accuracy may be necessary, such as integrating it with sensor-based solutions. In future work, we plan to extend the system to detect other risky driving behaviours, such as using a mobile phone or smoking, and further improve its accuracy to make it more suitable for real-world applications, closely approaching the accuracy achieved by sensor-based systems.

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